NIFTY Put–Call Parity Arbitrage under Real Market Frictions

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Contents

1 Introduction **2**
1.1 Methodology Overview . . . 2

2 Theoretical Framework **2**
2.1 Put Call Parity . . . 2
2.2 Assumptions of Put–Call Parity Equation . . . 3

3 Put–Call Parity under Market Frictions **3**
3.1 Practical Cost Breakdown . . . 4

4 Data Collection **5**

5 Analysis and Observation **5**
5.1 The Barrier of Transaction Friction . . . 5
5.2 Options Chain Microstructure . . . 6
5.3 Volatility Pulse and Temporal Persistence . . . 7
5.4 The Liquidity Filter and Directional Asymmetry . . . 8
5.5 The Time Value of Arbitrage: Expiry and Interest Rates . . . 9

6 Live Arbitrage Analysis System **10**

7 Limitations **11**

8 Conclusion **11**

9 Future Work **12**

10 References **12**

11 Contribution **13**

1 Introduction

Options pricing plays a crucial role in modern financial markets, providing insights into market expectations, risk management, and trading opportunities. One of the fundamental relationships governing option prices is the *Put–Call Parity*, which establishes a no-arbitrage condition between European call and put options on the same underlying asset with identical strike price and maturity.

In an ideal frictionless market, any deviation from this parity relation creates a risk-free arbitrage opportunity. However, real-world markets includes transaction costs, bid-ask spreads, margin requirements, and funding constraints, which significantly impacts the feasibility of such strategies.

The objective of this study is to bridge the gap between theoretical finance and real-world trading by assessing how often and to what extent arbitrage opportunities arise in a highly liquid market like NIFTY 50, and whether they can be effectively exploited by a retail trader and understand the role of cost and liquidity.

1.1 Methodology Overview

First, we start with the basic Put–Call Parity formula to see where arbitrage opportunities exist in theory.

Then, we make it realistic by adding all actual market costs like brokerage, taxes, bid–ask spread, funding cost, and margin requirements.

After that, we collected NIFTY 50 options data for 5 different expiries by scraping it from Groww.

Finally, we analyze the arbitrage opportunities by considering liquidity and all real market constraints to check whether they are actually profitable or not.

2 Theoretical Framework

2.1 Put Call Parity

We define the components of the parity equation as follows:

- C: Price of a European Call Option
- P: Price of a European Put Option
- S: Current Price of the Underlying Asset
- K: Strike Price
- r: Risk-free interest rate
- T: Time to expiration

The Put-Call Parity formula is:

$$C - P = S - Ke^{-rT}$$

To prove this, we analyze the payoffs at expiration (T). The payoff of a Call is $\max(S_T - K, 0)$ and the payoff of a Put is $\max(K - S_T, 0)$.

Step-by-Step LHS Calculation:

$$\text{LHS} = C_T - P_T$$

$$C_T - P_T = \max(S_T - K, 0) - \max(K - S_T, 0)$$

Case A: If $S_T > K$

$$C_T = S_T - K$$

$$P_T = 0$$

$$C_T - P_T = (S_T - K) - 0 = S_T - K$$

Case B: If $S_T \leq K$

$$C_T = 0$$

$$P_T = K - S_T$$

$$C_T - P_T = 0 - (K - S_T) = S_T - K$$

In both cases, the payoff is $S_T - K$. Since the RHS ($S_T - K$) matches the LHS payoff exactly, the portfolios must have the same value today.

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2.2 Assumptions of Put–Call Parity Equation

Below are few of assumption to hold put-call parity relationship which we will exploit:

- No transaction costs, taxes, or brokerage fees.
- No bid-ask spread (perfect liquidity).
- Constant and known risk-free interest rate.
- Unlimited borrowing and lending at the risk-free rate.

3 Put–Call Parity under Market Frictions

To move from theory to a practical retail trading system, we define the following friction variables (values taken from **Zerodha**, a leading brokerage platform for Indian retail traders, and publicly available financial sources):

- t_r: Turnover rate charged by broker

$$t_r = 0.03\% = 0.0003$$

- B: Flat brokerage cap per order

$$B = \ 20$$

  (Applied per executed order, not per quantity; for options, typically a flat 20 per trade)

- $\theta_c, \theta_p, \theta_s$: Transaction charge rates for Call, Put, and Stock respectively

$$\theta_c = 0.03553\%, \quad \theta_p = 0.03553\%, \quad \theta_s = 0.00307\%$$

- g: GST rate applied on brokerage and transaction charges

$$g = 18\% = 0.18$$

- $\tau_c^{sell}, \tau_p^{sell}, \tau_s$: STT/CTT rates

$$\tau_c^{sell} = 0.15\%, \quad \tau_p^{sell} = 0.15\%, \quad \tau_s = 0.025\%$$

  (STT is applied primarily on the sell side for options and on stock transactions)

- α, β: Margin requirements (SPAN + Exposure)

$$\alpha \approx 9.3\%, \quad \beta \approx 2\%$$

  - SPAN margin (risk-based): typically 9.3% for index derivatives (can go up to ∼17.7% for long expiries)
  - Exposure margin: approximately 2% of contract value for NIFTY options
  - Buy options: no margin (only premium paid)
  - Sell options: margin required = SPAN + Exposure − premium received

- r_b: Borrowing interest rate (cost of funds)

$$r_b \approx 11.5\%$$

  (Based on SBI personal loan rates: ∼10% + processing adjustments)

- r_l: Lending interest rate (return on idle capital)

$$r_l \approx 6.6\%$$

  (Based on SBI fixed deposit rates)

- Δ_{spread}: Bid-ask spread loss across all legs

$$\Delta_{spread} \approx 0 \quad \text{(assumed negligible or modeled separately)}$$

- N: Lot size (contract multiplier)

$$N = 65 \quad \text{(NIFTY)}$$

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3.1 Practical Cost Breakdown

A. Brokerage (Brk): Computed individually for the three legs because of the flat-fee cap:

$$\text{Brk} = \min(t_r \cdot C, B) + \min(t_r \cdot P, B) + \min(t_r \cdot S, B)$$

B. Transaction Charges (Txn):

$$\text{Txn} = \theta_c C + \theta_p P + \theta_s S$$

C. Government Taxes:

$$\text{GST} = g \cdot (\text{Brk} + \text{Txn})$$

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4. Case 1: Conversion (C - P is Overvalued)

Condition: $C - P > S - Ke^{-r_l T}$

Execution Steps:

1. Sell Call ($+C$)
2. Buy Put ($-P$)
3. Buy Stock ($-S$)
4. Borrow $Ke^{-r_l T}$ at lending rate logic ($+Ke^{-r_l T}$)

Initial Cashflow (Π_0):

$$\Pi_0 = C - P - S + Ke^{-r_l T}$$

Total Frictional Cost ($Cost_1$):

$$Cost_1 = \text{Brk} + \text{Txn} + \text{GST} + (\tau_c^{sell} C + \tau_s S) + \Delta_{spread} + K(e^{-r_l T} - e^{-r_b T}) + [(\alpha + \beta)S - C](e^{r_b T} - 1)$$

Profit Requirement:

$$\Pi_0 > Cost_1$$

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5. Case 2: Reversal (C - P is Undervalued)

Condition: $C - P < S - Ke^{-r_b T}$

Execution Steps:

1. Buy Call ($-C$)
2. Sell Put ($+P$)
3. Sell Stock ($+S$)
4. Lend/Deploy $Ke^{-r_b T}$ ($-Ke^{-r_b T}$)

Initial Cashflow (Π_0):

$$\Pi_0 = S - C + P - Ke^{-r_b T}$$

Total Frictional Cost ($Cost_2$):

$$Cost_2 = \text{Brk} + \text{Txn} + \text{GST} + (\tau_p^{sell} P + \tau_s S) + \Delta_{spread} + K(e^{-r_b T} - e^{-r_l T}) + [(\alpha + \beta)S - P](e^{r_b T} - 1)$$

Profit Requirement:

$$\Pi_0 > Cost_2$$

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6. Final Scaled Inequality

To check for a tradeable "Effective Violation" for 1 lot:

$$\text{Net Profit} = N \times (\text{Violation Amount} - \text{Total Costs})$$

The final trade signal triggers only if:

$$\boxed{\text{Effective Violation} > \sum(\text{Brk} + \text{Txn} + \text{GST} + \text{STT} + \text{Spread} + \text{Funding for rate mismatch } r_b \neq r_l. + \text{Margin Cost})}$$

4 Data Collection

To analyze real-world arbitrage opportunities, we collected NIFTY 50 options chain data directly from Groww. The data includes both Call and Put options across multiple strike prices along with the corresponding spot price, option prices (LTP), open interest, and Greeks.

The data was collected for five different expiries to capture both near-term and long-term market behavior:

- April 7, 2026 (expiry day and the day before expiry – 2 days of data)
- April 28, 2026
- June 30, 2026
- September 29, 2026
- December 29, 2026
- Total data points/rows ≈ 5.5 lakh (500k)

For each expiry, data was collected continuously during market hours. Although the collection script was run at a frequency of 2 seconds, the effective update frequency of the data was approximately 30–32 seconds due to the refresh rate of the Groww interface. As a result, intermediate data points were often duplicated.

Each data entry contains a timestamp (`fetch_time`), along with option-specific details such as strike price, Call and Put LTPs, open interest, and Greeks. This structured format allows us to compute Put–Call Parity violations and further adjust them for real-world trading costs.

Overall, the dataset provides a time-series view of the option chain, enabling analysis of arbitrage opportunities across different strikes, expiries, and market conditions while accounting for liquidity and execution constraints.

5 Analysis and Observation

This section provides a multi-dimensional evaluation of the Put-Call Parity (PCP) relationship within the Nifty 50 index. By analyzing the data through ten distinct lenses, we transition from theoretical price deviations to realizable market alpha, identifying the structural barriers that define the Indian derivatives market.

5.1 The Barrier of Transaction Friction

The primary inhibitor of arbitrage is the "Cost Stack." We examine how gross violations are eroded by regulatory and execution costs.

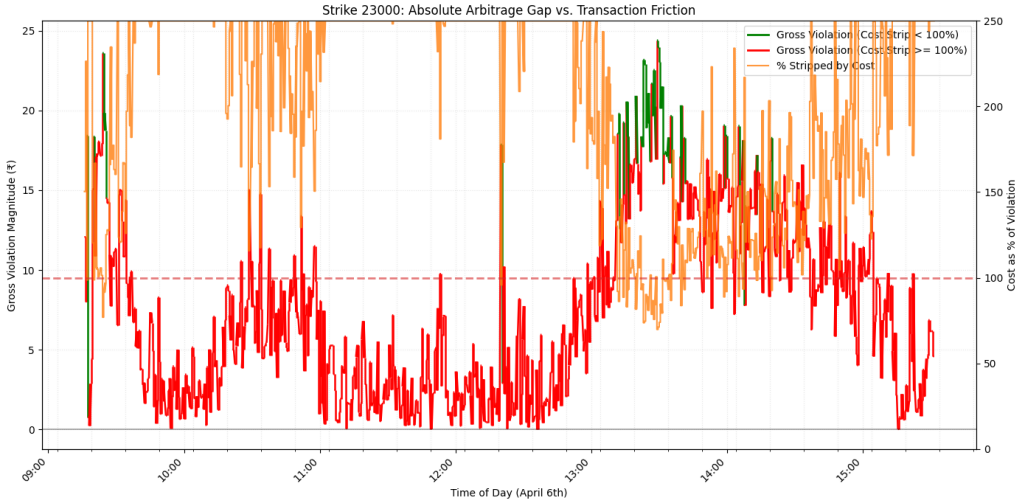

Figure 1: Arbitrage vs. Friction: The transition from Gross Violation to Net Profit.

Economic Observation: Figure 1 illustrates the **No-Arbitrage Band**. The market is not "perfectly" efficient; rather, it is **Economically Efficient**. Deviations exist, but they are often smaller than the cost of execution.

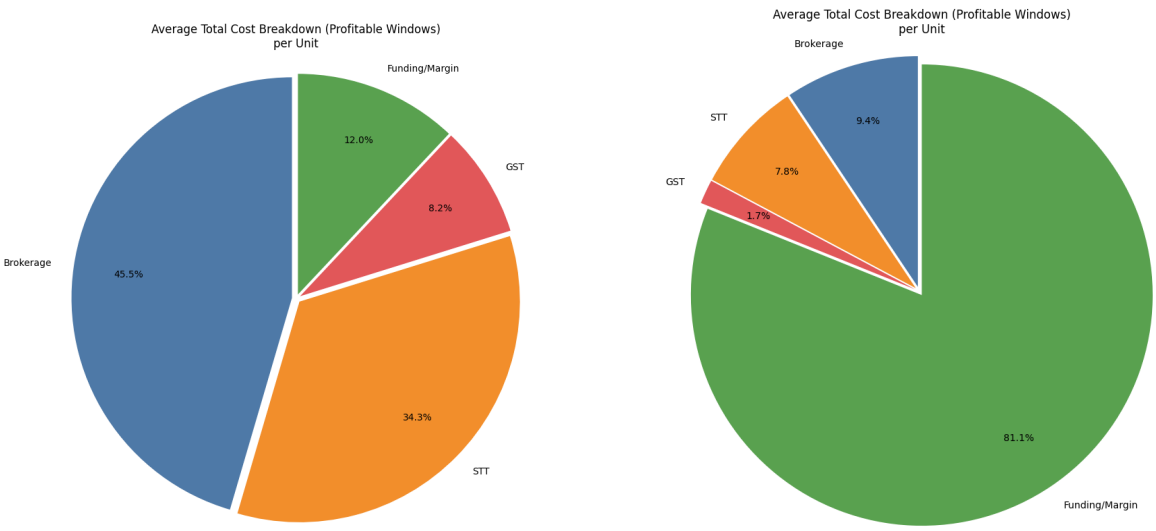

Figure 2: Cost Breakdown Comparison: (Left) Near Expiry (day of expiry and day before expiry) where funding cost is negligible ($t \to 0$), (Right) Longer Expiry (4 to 8 months) where funding cost becomes significant.

Economic Justification: As time to expiry approaches zero ($t \to 0$), the **funding cost becomes negligible**, leaving transaction costs such as **STT, GST, and Brokerage** as the dominant factors. Regulatory friction—especially **brokerage on the sell-side**—acts as a "gatekeeper," ensuring that only high-magnitude arbitrage opportunities are exploited while smaller discrepancies persist as untradable noise.

For **longer expiries**, however, **funding cost becomes a major component**, effectively acting as the cumulative outcome of all cost factors and significantly influencing arbitrage feasibility.

5.2 Options Chain Microstructure

We analyze the "Moneyness" of violations to determine which segments of the option chain are most prone to mispricing.

Figure 3: Intraday Heatmap of Net Profit Opportunities across Strike Prices.

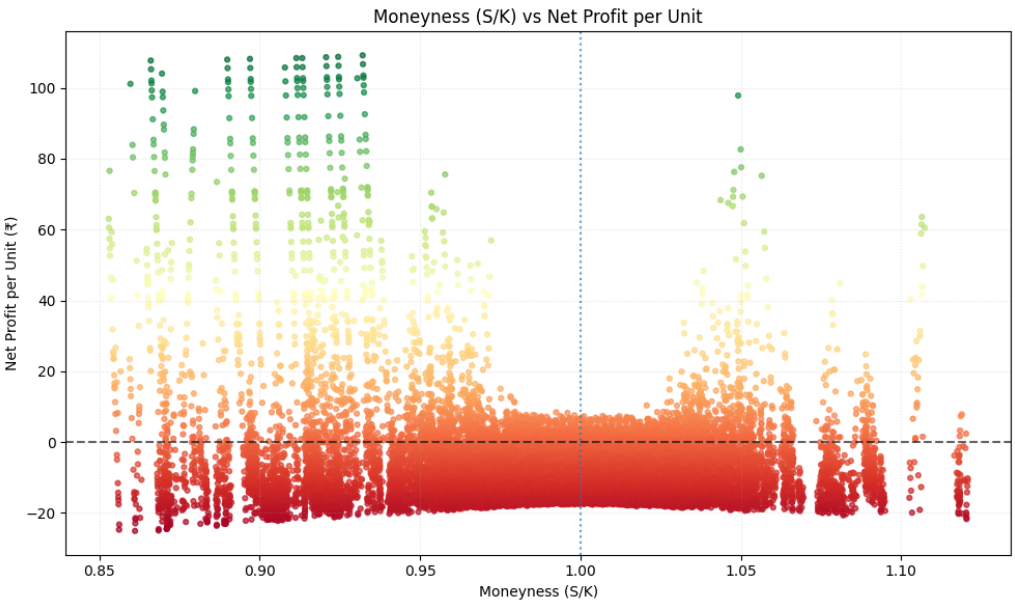

Figure 4: Profitability across Moneyness (S/K Ratio).

Economic Justification: Profitability follows a **"Moneyness Smile."** At-the-Money (ATM) strikes (S/K $\approx$ 1.0) exhibit near-perfect efficiency due to intense competition from High-Frequency Trading (HFT) firms. Conversely, the "wings" (Deep ITM/OTM) show larger violations due to lower liquidity and wider bid-ask spreads.

5.3 Volatility Pulse and Temporal Persistence

Arbitrage is a transient phenomenon, often triggered by sudden liquidity shifts or global market events.

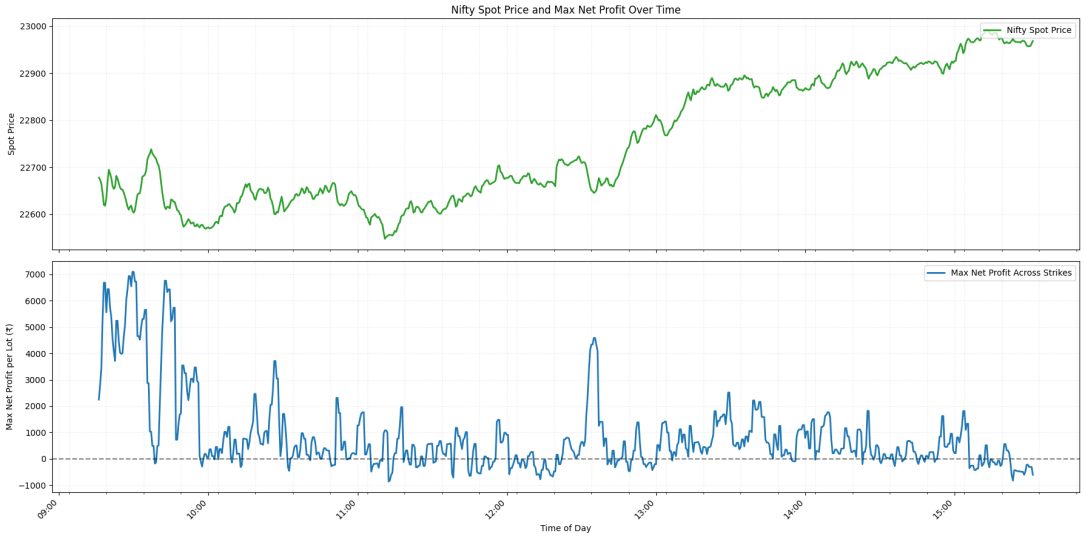

Figure 5: Correlation between Nifty 50 Spot Price and Realizable Net Profit.

Economic Justification: The most significant spikes in realizable net profit are observed during the Market Opening (09:15 – 10:00 IST) and again shortly before 12:30 IST, driven by distinct liquidity events. The initial morning pulse represents the market's reaction to overnight global news and the price discovery process, where high volatility leads to temporary mispricing between Nifty spot and derivative contracts. The subsequent spike occurring before 12:30 IST reflects institutional pre-market positioning ahead of the European Open; as global traders interact with Indian derivatives like GIFT Nifty, local prices momentarily lag behind spot movements, creating a "Volatility Pulse" that breaks parity and provides a brief window for higher realizable profit.

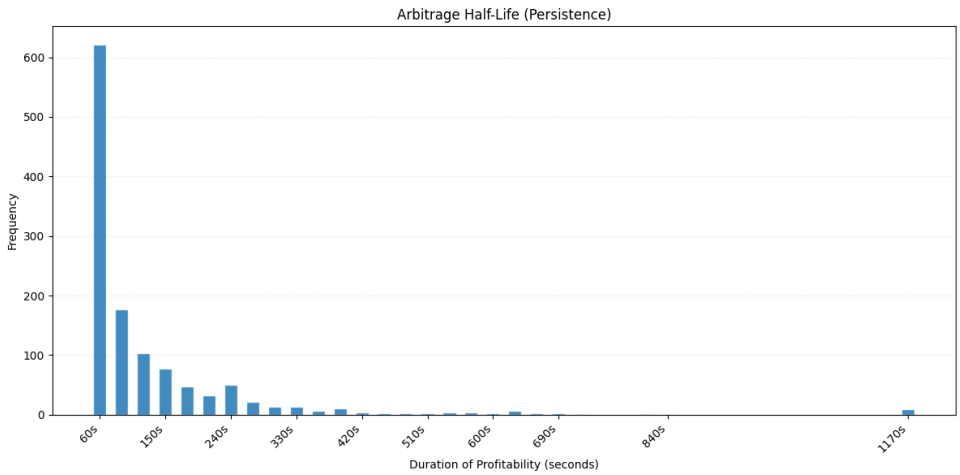

Figure 6: Arbitrage Half-Life: Frequency of consecutive profitable intervals.

Economic Justification: Information Decay. Figure 8 shows that 90% of signals vanish within 60 seconds. This "self-correction" speed proves that the Nifty 50 is a fast-updating market; any signal lasting longer than 5 minutes is likely a "data ghost" from an illiquid, stale price.

5.4 The Liquidity Filter and Directional Asymmetry

Theoretical profits are often "Ghost Profits" that exist only in illiquid strikes.

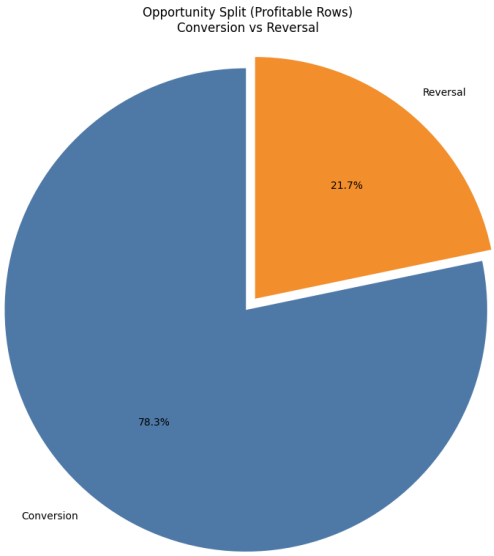

Figure 7: Reversal vs. Conversion (Count)

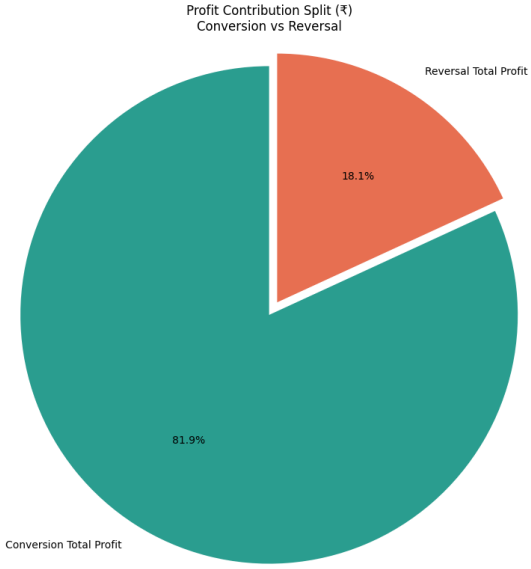

Figure 8: Reversal vs. Conversion (Rupees)

Economic Observation: Short-Selling Constraints. Reversals (overpriced Puts) are rarer than Conversions. This is because the "Short-Spot" leg of a reversal is difficult and capital-intensive for many retail participants, leaving Put mispricings open longer than Call mispricings.

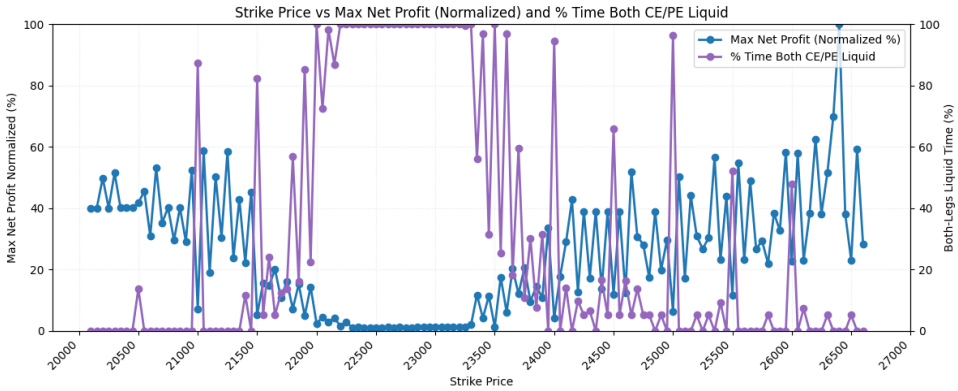

Figure 9: Arbitrage Violation Magnitude vs. Strike Liquidity.

Economic Justification: Adverse Selection. We observe that the largest violations occur in illiquid strikes. These represent untradable "paper alpha" where the Last Traded Price (LTP) is stale and does not reflect current market reality.

5.5 The Time Value of Arbitrage: Expiry and Interest Rates

The "Time" component (T) of the PCP equation determines the sensitivity of the arbitrage.

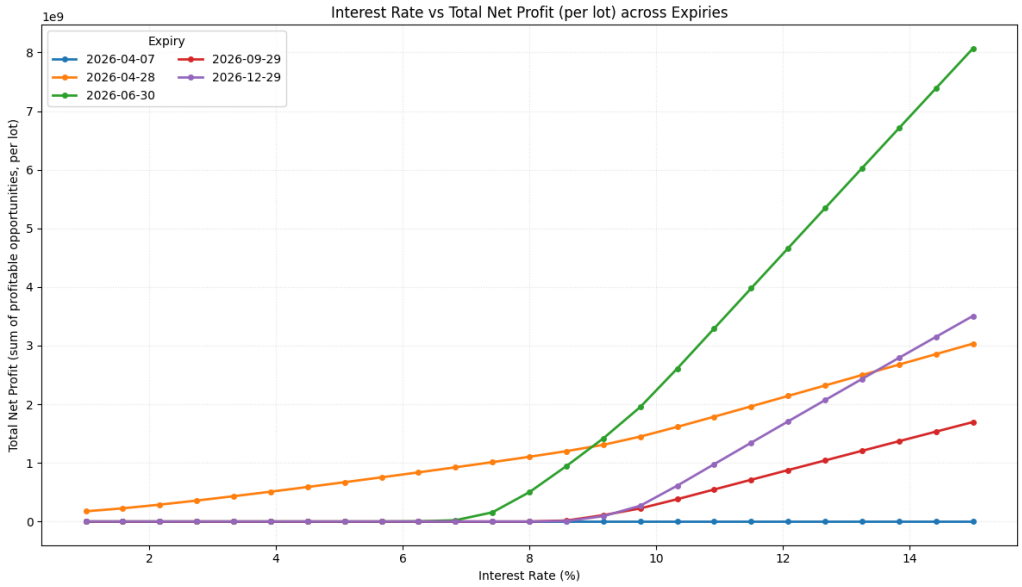

Figure 10: Interest Rate Sensitivity (r-Volatility) for Near vs. Far Expiries.

Economic Observation: The Duration Effect. In far-dated contracts (June/December), the cost of carry (borrowing at 11.5%) becomes the dominant factor. Small shifts in r move the parity price more than the actual market mispricing.

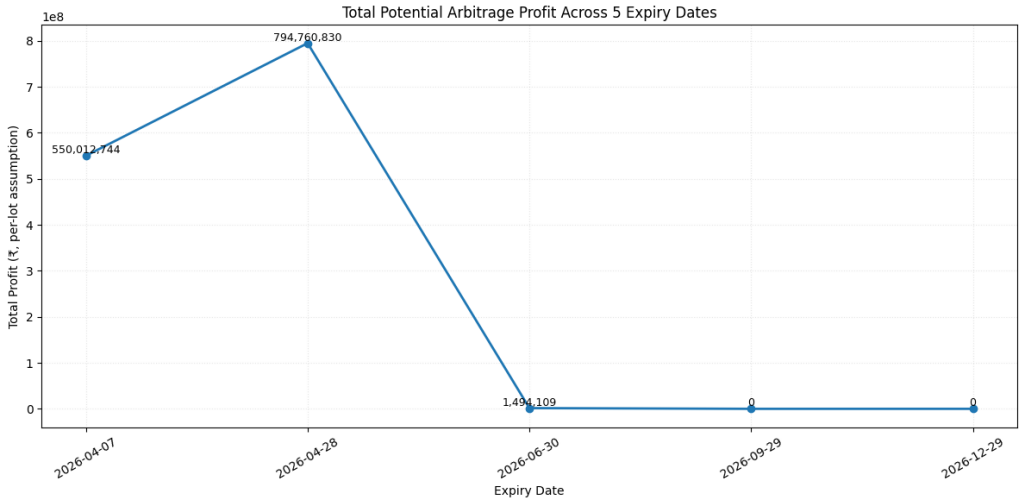

Figure 11: Total Potential Arbitrage Profit across 5 Expiry Dates.

Economic Justification: Economic Justification: Profit potential peaks at the Monthly Expiry (April 28) rather than the nearest weekly expiry, highlighting the role of Liquidity-Induced Efficiency. While $T \to 0$ generally increases arbitrage viability, the April 28 contract benefits from significantly higher Open Interest and Volume, which reduces execution slippage and allows for the capture of smaller price-parity deviations. Additionally, the Monthly Roll-Over period introduces institutional flow imbalances that create more frequent "Volatility Pulses" compared to the thinner weekly contracts, resulting in a higher aggregate potential profit despite a slightly longer time to maturity.

6 Live Arbitrage Analysis System

We also built a simple live dashboard to track arbitrage opportunities in real time. **Live App:** `https://nifty50-arbitrage-detection.streamlit.app/`

- Built using Python, Streamlit (UI), Plotly (visualization), and Pandas (data handling)

- Fetches live NIFTY option chain data from Groww for a given expiry
- Computes Put–Call Parity violation for all strike prices
- Adjusts the violation for real costs (brokerage, taxes, funding, margin)
- Shows arbitrage as % of spot price for easy comparison
- Displays liquidity using open interest (CE vs PE)
- Helps filter out "fake" opportunities which are not tradable due to low liquidity
- Auto-refreshes every ∼30 seconds to capture short-lived opportunities

This tool helps move from theoretical arbitrage to what is actually possible in real markets.

- A detailed guide and the complete code for the bot are available at:
  `https://github.com/rahulkhichar7/Nifty50-PutCallParity-Arbitrage`

7 Limitations

- Currently, we assume zero cost for slippage and delayed execution. This can be incorporated by modifying the `spread_cost` parameter in the `config.py` file.
- NSE does not allow direct shorting of the index; futures must be used instead, introducing basis risk.
- Data latency of approximately 32 seconds may result in missed or stale arbitrage signals.
- Past performance and results based on the current analysis do not guarantee future outcomes, as the study has been conducted on a limited dataset.

8 Conclusion

- From the analysis of the graphs and observations, it is clear that Put–Call Parity violations occur in the NIFTY options market. At multiple points the theoretical relationship between call, put, and spot prices is not perfectly maintained.
- But when the real world factors are considered, such as brokerage, transaction charges, GST, STT, and funding costs, most of these violations are reduced significantly. The graphs show that while gross arbitrage opportunities may appear large, the net profit after costs is often close to zero or negative.
- Liquidity plays an important role in determining whether an arbitrage opportunity can be executed or not. The results indicate that profitable opportunities are concentrated in illiquid strikes. These are not reliable and cannot be actually traded.
- Arbitrage opportunities are extremely short-lived, with most of them disappearing within seconds due to rapid market correction. This highlights the dominance of high-frequency trading and fast information flow in maintaining the market efficiency.
- We can also find a directional asymmetry, where conversion opportunities are more common than reversal opportunities due to constraints in short-selling the underlying asset. Expiry plays a significant role, with arbitrage being more feasible near expiration when funding costs are minimal, whereas long-dated contracts are largely protected from arbitrage due to the cost of carry
- Overall, the findings suggest that true arbitrage opportunities for retail traders are rare, small, and difficult to capture. Most observed violations are either limited by liquidity, transaction costs or eliminated by rapid market adjustments. Therefore, while theoretical arbitrage exists, practical arbitrage in real markets is constrained and highly competitive.
- **Disclaimer**: This tool is for educational and research purposes only. Trading options involves substantial risk. Always conduct your own due diligence and consult with financial advisors before trading. Past performance and analysis do not guarantee future results.

9 Future Work

- Use higher-frequency and lower-latency data (tick-level or real-time feeds) to improve precision and reduce stale signals.
- Incorporate bid–ask prices instead of LTP to model realistic execution and reduce false arbitrage signals.
- Develop an automated trading bot to detect opportunities and suggest appropriate actions, integrated with the Zerodha API for real-time order execution.
- Extend the framework to include index futures for better replication of short positions and reduced basis mismatch.
- Include multiple assets and optimize the portfolio for arbitrage.
- Introduce machine learning models to predict persistence of arbitrage opportunities and filter noise.

10 References

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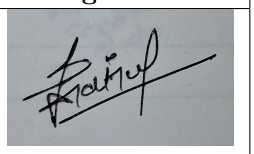
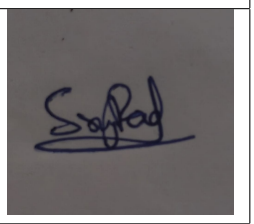
- https://github.com/rahulkhichar7/Nifty50-PutCallParity-Arbitrage

11 Contribution

Name	Contribution	Signature
Rahul Khichar	Project planning, Derived Effective violation criteria, Collected Data, web app for live analysis (Section 1,2,3,4,6,7 & 9 in report and prepared slides for these sections.)	
Jils Shah	Project Analysis, Data Collection	
Padala Srujan	worked on report	
Shivangi Thaker	helped with analysis and made the slides for presentation	